THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO SECONDARY SCHOOL

FORM TWO PRE-NATIONAL EXAMINATION 1
033 BIOLOGY
MARKING SCHEME APRIL 2026

SECTION A (15 MARKS)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	B	D	C	A	B	D	C	C	C

2.

(i)	(ii)	(iii)	(iv)	(v)
B	A	D	C	E

SECTION B (70 MARKS)

3.

(i) Eye piece lens (Ocular lens): it is used to magnify specimen after having been magnified by the objective lens.
(ii) Body tube: supports the eyepiece and the revolving nose piece.
(iii)Revolving nose piece: it supports the objective lenses.
(iv)Objective lens: - It is used for directly magnifying specimen. (02 @ = 10 marks)
(v) Stage: is the place where the slide with specimen to be observed is placed.

4. (a)

(i) Binomial Nomenclature: This is the formal system of naming species of living things by giving each a name composed of two parts, both of which use Latin grammatical forms: the **Genus** name and the **Species** name. (02 marks)

(ii) Rules:

a) The Genus name must start with a capital letter, while the Species name starts with a small letter.
b) When handwritten, the two names must be underlined separately; when printed, they should be in *italics*.
c) The names are usually in Latin or Latinized. (01½ @ = 04½ marks)

(b)

(i) **Class:** Mammal (01½ marks)
(ii) **Kingdom:** Animalia. **Phylum:** Chordata. (02 marks)

5. (a)

(i) Red blood cells: facilitates transportation of oxygen.

(ii) White blood cells: help to resist infections and parasites.
(iii) Platelets (thrombocytes) are important in the clotting of blood.
(iv) Blood plasma: maintaining blood pressure/volume, aiding clotting, regulating pH balance, ... (02 @ = 08 marks)

(b) The **left ventricle** has thicker walls because it must pump oxygenated blood with high pressure to the rest of the entire body, unlike the right ventricle which only pumps blood to the lungs. (02 marks)

6. Factors affecting rate of gas exchange:
   (i) Physical activities: A more active body requires more oxygen than a less active body, therefore the rate is higher when a person is more active.
   (ii) Concentration of hemoglobin: when a person has a low concentration of haemoglobin, such as an anemic person, the person's rate of gas exchange increases.
   (iii) Age: the rate of gas exchange is higher in young people than old people since young people are more active and growth process still takes place in young people.
   (iv) Carbon dioxide concentration: in an environment with high carbon dioxide concentration, the rate of gas exchange of an individual increases.
   (v) Health status: the rate is higher in a sick person due to increased metabolism (by the liver) (02 @ = 10 marks)
   (vi) Altitude: the rate is higher in high altitudes due to decrease in atmospheric pressure and oxygen concentration.

7. (a)
   (i) Prokaryotic: cells do not have membrane bound organelles, example; bacteria is a prokaryotic organism. Eukaryotic: cells have membrane bound organelles, for example, amoeba, plants and animals are eukaryotic organisms.
   (ii) Prokaryotic cells are smaller and lack true nucleus, where as Eukaryotic cells are larger and have true nucleus (02 @ = 04 marks)

   (b)
   (i) Cell differentiation is the presence of different types of cells which are specialized to perform different functions. Importance: division of labour which helps the body to carry out its activities more efficiently. (04 marks)
   (ii) types of <u>cells</u> in plants are the <u>root hair, xylem</u> and <u>phloem</u> , and in animals are the <u>sperm</u>, <u>blood</u> and <u>nerve cells</u>. (00½ @ = 02 marks)

8. (a)
   (i) They cause decomposition of matter hence releases nutrients to plants.
   (ii) Some aid in digestion, example the bacteria in the gut of the ruminants.
   (iii) Some produce vitamins K and B_{12} in the human gut.
   (iv) Some convert atmospheric nitrogen to nitrates for plant use.
   (v) Some are used in the production of antibiotics.
   (vi) Some are used in making vaccines.
   (vii) Some are used in sewage treatment (They decompose pollutants).

(viii) Some are used in the food industry for cheese production, yogurt and sauerkraut production.

(ix) Some are used in genetic engineering to produce insulin.

(x) Autotrophic bacteria release oxygen in the atmosphere.

(xi) Some are used in biological researches. (Any five.., 2@ = 10 marks)

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9. (a) First test for reducing sugar as follows:

Put a solution of the substance to be tested in a test tube and add Benedict's solution (blue) and heat gently. If reducing sugar is present, a color change from blue, through green to yellow and orange/red shows will be observed. If the Benedict's test comes out negative, test disaccharides (non-reducing sugars) as follows *03 marks*

Add dilute hydrochloric acid to the food sample solution and then heat for about one minute (to split up the non-reducing sugar into reducing sugar). After cooling, add an equal amount of dilute sodium hydroxide solution or sodium hydrogen carbonate solution (to neutralize the acid) and then proceed with the Benedict's test. Positive results for Benedict's test indicate the presence of non-reducing sugar originally in the food sample. *04 marks*

(b) Run an Iodine test: Add a couple of drops of iodine solution (brown) directly to the substance to be tested, some food substances may have to be cooked /boiled before testing. A blue-black coloration indicates the presence of starch in the food. *03 marks*

SECTION C (15 MARKS)

10. The adaptations of the parts of the heart to their functions

- The heart has muscles that are never fatigued.
- The heart has Sino Atrial Node that act as a pacemaker by regulating the heartbeat rates and excitation of the heart.
- There are coronary arteries which lay over the heart surface to provide the heart muscle itself with oxygenated blood.
- Has muscular ventricles, with the left ventricle being more muscular to facilitate pumping of the oxygenated blood around the body.
- There are tricuspid and bicuspid valves between the atria and the ventricles, which ensure the blood flows upwards into the major arteries and not back into the atria.
- At the base of the major arteries, there are semilunar valves which prevent the blood returning to the heart as the ventricles relax. Any $5 \times 03 = 15$ marks
- There is pericardium (pericardial membrane) around the heart that produces lubricating fluid and supports the heart.

Total: 15 marks